Some Unpleasant New Keynesian Arithmetics

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Abstract

This research proposal aims to bridge the divide between standard New Keynesian analysis of monetary-fiscal interactions and the one yielded by the fiscal theory of the price level (FTPL). It achieves this synthesis by taking inspiration from Sargent and Wallace's seminal *Some Unpleasant Monetarist Arithmetic* in two ways: in spirit, we present a minimalist New Keynesian model that adheres to all of the main orthodox tenets. In method, the key addition to the model is adding agent heterogeneity via a OLG with financial segmentation module. We expect to find richer and more credible channels though which monetary policy, fiscal policy, prices and output interact.

1 Introduction

Recent inflationary episodes throughout the developed world have revived the debate over the fiscal roots of inflation. One of the main thrusts of attack to this question has been the literature on the fiscal theory of price level (FTPL) as it manifests itself on the standard New Keynesian model. This paradigm features a cashless economy, where the only role of money is that of *numéraire*. Agents hold no money balances, and the exchange, savings and liquidity motives are absent.

This is an inherited feature of the models developed by Clarida and Gertler (1999), Woodford (2003) and others, where the economy is taken to the "cashless limit". Because of this, the entire modern literature is somewhat disconnected from its intellectual forefathers, in either Keynesian models where governments pursue seigniorage while disliking inflation, à lá Cagan (1956), or microfounded monetarist ones where inflation arises from the savings motive as in Wallace and Sargent (1981). We want to bring attention to this second tradition.

Sargent' and Wallace's insight, and the significance of their work, was due to how their counterintuitive results obtained even though they adhered closely to what was then the orthodoxy in macroeconomic thought. That from a strictly monetarist economy made off of optimizing agents there could arise an economic system where monetary policy held no sway over inflation was astonishing. This methodological approach of keeping to not only the letter, but also the spirit of the prevalent mode of thought is mostly absent from the current FTPL literature. Instead, in the models first proposed by Leeper (1991) and summarized by Cochrane (2021) key assumptions of standard NK models are challenged or disregarded: governments do not attempt to stabilize deficits, and central banks do not adhere to the Taylor principle.

Proponents of the theory argue that this is a feature, not a bug, and a closer representation of how the world actually works (for a recent example, see Cochrane (2025)). It nonetheless creates a disconnect between the standard, or orthodox, way of thinking about inflation and monetary-fiscal interactions, and what is defended by FTPL.

We seek to bridge this disconnect by following Sargent and Wallace's *Some Unpleasant Monetarist Arithmetic* in both method and mean: in method, by altering as few of the fundamentals about how the standard New Keynesian model operates as possible, and in mean by introducing the same microfounded novelty, namely substituting the representative agent by a financially segmented OLG block.

This structure also puts us in contact with the recently developing literature on heterogeneous agents in New Keynesian framekworks (HANK). While financial segmentation as a means to heterogeneity is already common in the New Keynesian tradition (as seen in the numerous two-agent models featuring hand-to-mouth agents), only recently has the OLG route been re-found, as in Angeletos and Wolf (2026) and Rachel and Ravn. (2026). Among modern studies on monetary-fiscal interactions these are the closest correlates to the present work: we share partially both the instrument by which we differentiate from previous literature, and the intention of mending the methodological break that divides standard New Keynesian models from their FTPL cousins.

2 Proposed Model

The model we propose is mostly textbook New Keynesian: firms are in Dixit-Stiglitz monopolistic competition, prices are updated under a Calvo process, the monetary authority follows a policy rule, which we will specify to adhere to the Taylor principle, and the fiscal authority follows a debt balancing rule which will not fully finance deficit shocks. Labor markets are perfectly competitive. Since we do not wish to study business cycle or monetary shocks, the single possible shock to the system is the fiscal one.

The key difference is in the agent, which will be ported from Sargent' and Wallace's second model in *Some Unpleasant Monetarist Arithmetic*. We will thus have two classes of agents inste4ad of a single representative one: "financially illiterate", forbidden from buying government debt, and "financially literate", which may do so (henceforth abbreviated as I and L respectively). Each agent lives and works over two periods, following the perpetual youth tradition. At any given period after the first one, then, we have four different types of agent: I1, I2, L1 and L2. Age-1 agents are more productive than age-2 ones, thus earning a higher wage. Agent age is public information.

Type-I agents may save in the form of a non-interest bearing asset (money); usually in NK setups agents that can only save like this choose not to hold any savings whatsoever. We expect that the decaying productivity of labor in agent age might introduce a savings motive strong enough to generate positive money balances, as agents want to smooth consumption but earn higher wages on the first period of life. If that is the case, we would successfully replicate Sargent and Wallace's "poor" agent (the one that is prohibitted from buying government debt under their model) under a OLG-NG framework, and be able to derive a money demand schedule without resorting to adding money directly to agents utility as in Gali (2008).

In other OLG-NK setups (Angeletos and Wolf (2026), Rachel and Ravn. (2026)), agents die off randomly, which leads to the inclusion of unwieldy inheritance-redistributing "social funds" for transferring the savings of recently-deceased agents to recently-born ones. One of the advantages we expect to develop is the doing away with this kind of mechanism by eliminating the random risk of death. In our setup, agents always live two periods, and fully consume their savings on the second and last period of life.

This model should bring forth several differences in relation to the standard New Keynesian ones. As is common with HANK models, we expect to introduce private debt holdings into the IS curve. Agents will see government debt as wealth, breaking with Ricardian equivalence and allowing fiscal policy and inflation to directly interact with the IS. Furthermore, we believe this setup will reintroduce the savings motive of money, generating another channel through which inflation affects the IS curve.

3 Expected Findings

We expect to find a richer monetary-fiscal interaction surface. Rather than the simple standard NK, or the controversial FTPL channels, this framework should produce less direct, more ambiguous results for how inflation, output, and government debt respond to fiscal shocks. We also expect these results to be more intuitive and more in line with what is seen in the real world.

In addition, we would hope to reproduce both of the original "unpleasant results" found by Sargent and Wallace: that in some cases, more restrictive monetary policy leads to lower inflation today at the cost of higher inflation tomorrow, and that in extreme cases it leads to higher inflation both today and tomorrow.

Lastly, this modeling strategy could become an additional instrument in the expanding toolkit of HANK models. It would have the advantage of being more interpretable and less convoluted than other OLG setups, while still being more tractable than those in the style of Auclert (2019) and Kaplan and Violante (2018), as it allows for lightweight simulations and closed-form results. This would configure a step towards a new benchmark model of monetary and fiscal policy for the short run that would combine both recent and seminal contributions.

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